

Section two

- 1.8 in the book

An analog electrocardiogram "ECG" signal contains useful frequencies up to 100 Hz

a. What's the Nyquist rate for this signal?

b. Suppose that we sample this signal at a rate of 250 samples.

What's the highest frequency that can be represented uniquely at this sampling rate?

Answer:

a. Nyquist $F_s \geq 2 F_{max}$

Since, $F_{max} = 100 \text{ Hz}$ "Given in the problem"

Therefore, $F_s \geq 2(100) \rightarrow F_s = 200 \text{ Hz}$

"According to the sampling theorem"

b. Highest frequency that can be represented uniquely = "Folding Frequency"

"The frequency $F_s/2$ "or $\omega = \pi$ " is called "Folding Frequency"

$$F_{\text{fold}} = \frac{F_s}{2} = \frac{250}{2} = 125 \text{ Hz}$$

-1.9 in the book

-An analog signal $x_a(t) = \sin(480\pi t) + 3\sin(720\pi t)$ is sampled 600 times per second

- Determine the Nyquist Sampling rate at $x_a(t)$
- Determine the Folding Frequency
- What are the frequencies in radian, in the resulting discrete time signal $x(n)$?
- If Can we Construct again "reconstruct" this signal again correctly from the sampled signal using the previously mentioned Sampling Frequency.

Answer:

a. Nyquist Rate = $F_s \geq 2F_{\max}$

→ There exists above two frequencies

$$F_1 = 480/2 = 240$$

$$F_2 = 720/2 = 360$$

So, Maximum Frequency = 360

$$\text{And, } F_s \geq 2(360) \rightarrow F_s = 720$$

"According to : $x_a(t) = A \cdot \cos(2\pi F t + \theta)$

b. Folding Frequency " F_{fold} " = $\frac{F_s}{2} = \frac{600}{2} = 300 \text{ Hz}$ → Sampling Frequency

C- $x_c(n) = x_a(n.T)$ Discrete

↳ Sampling Interval

Since, $F_s = \frac{1}{T}$ Sampling Frequency

So, $x(n) = x_a\left(\frac{n}{F_s}\right)$

$$= \sin\left(\frac{480\pi n}{600}\right) + 3 \cdot \sin\left(\frac{720\pi n}{600}\right)$$

$$x(n) = \sin\left(\frac{4\pi n}{5}\right) + 3 \cdot \sin\left(\frac{6\pi n}{5}\right)$$

$$= \sin\left(\frac{4\pi n}{5}\right) + 3 \cdot \sin\left(2\pi - \frac{4\pi n}{5}\right)$$

$$= \sin\left(\frac{4\pi n}{5}\right) + 3 \cdot \sin\left(-\frac{4\pi n}{5}\right)$$

$$= \sin\left(\frac{4\pi n}{5}\right) - 3 \cdot \sin\left(\frac{4\pi n}{5}\right) = -2 \sin\left(\frac{4\pi n}{5}\right)$$

Therefore: $\omega = \frac{4\pi}{5}$

d. No, we can't due to the aliasing happened to the second component using the considered Sampling Frequency.

1.10 in the book.

A digital Communication link carries binary. Coded words representing Samples of an input Signal.

$$x_a(t) = 3 \cos 600\pi t + 2 \cos 1800\pi t$$

The link is operated at 10,000 bits/s and each input sample is quantized into 1024 diff. voltage levels

- What is the Sampling Freq. and Folding Freq.?
- What is the Nyquist rate for the signal $x_a(t)$?
- What are the frequencies in the resulting discrete-time signal $x(n)$?
- What is the resolution Δ ?

Answer:

a. Number of bits/sample = $\log_2 1024 = 10$

Since, 10,000 is number of bits/s

Then: $F_s = \frac{\text{No. of bits/s}}{\text{No. of bits/sample}} = \frac{10,000}{10} = 1000 \text{ Sample/sec}$

- Folding Frequency " f_{fold} " = $\frac{F_s}{2} = \frac{1000}{2} = 500 \text{ Hz}$

b. We have 2 Frequencies: $F_1 = \frac{600}{2} = 300$, $F_2 = \frac{1800}{2} = 900$

So, Maximum Frequency = 900 Hz

Then, Nyquist Frequency = $2 \times \overset{\text{Max. Frequency}}{900} = 1800 \text{ Hz}$

c. Relative or Normalized Frequency " f " = $\frac{F}{F_s}$

Since, $x(n)$ has two frequencies

Then, we should compute Normalized Frequency for both of them.

$$f_1 = \frac{F_1}{F_s} = \left(\frac{600}{2}\right) * \left(\frac{1}{1000}\right) = 0.3$$

$$f_2 = \frac{F_2}{F_s} = \left(\frac{1800}{2}\right) * \left(\frac{1}{1000}\right) = 0.9 \xrightarrow{2\pi(1) - 2\pi(0.1) = 2\pi(0.1)} > 0.5 \Rightarrow f_2 = 0.1$$

"From properties of $f \rightarrow -\frac{1}{2} < f < 0.5$ ".

$$\text{Therefore: } x(n) = 3 \cdot \cos[(2\pi)(0.3)n] + 2 \cdot \cos[(2\pi)(0.1)n]$$

$$\text{"In general: } x(n) = A \cdot \cos\left(\frac{2\pi n F}{F_s} + \theta\right)"$$

$$d. \Delta = \frac{(x_{\max} - x_{\min})}{L}$$

$x_{\max}$: Maximum Amplitude

$x_{\min}$: Minimum Amplitude

L : No. of levels "zones" "1024 is given"

$$x_{\max} = 5 \text{ as } 3(1) + 2(1) \text{ will be the max. value}$$

$$x_{\min} = -5$$

$$\text{So, } \Delta = \frac{5 - (-5)}{1024} = \frac{10}{1024}$$

4) Quantize and encode the following sampled signal $x(n)$ using 4 quantization levels and minimum number of bits. Compute the average error power in both cases

$$x(n) = \{-1.22, 1.5, 3.24, 3.94, 2.20, -1.10, -2.26, -1.88, -1.2\}$$

Maximum Amplitude = 3.94

Minimum Amplitude = -2.26

$$\Delta = \frac{\text{Max. Amplitude} - \text{Min. Amplitude}}{L}$$

$$= \frac{3.94 - (-2.26)}{4} = 1.55$$

4
↳ Given

* The Four levels are:

$$[-2.26, -0.71]^{(1)}, [-0.71, 0.84]^{(2)}, [0.84, 2.39]^{(3)}, [2.39, 3.94]^{(4)}$$

* The midpoints of the Four levels:

$$\{-1.485^{(1)}, 0.065^{(2)}, 1.615^{(3)}, 3.165^{(4)}\}$$

n	$x(n)$	Interval Index	$x_q(n)$	$e_q(n) = x_q(n) - x(n)$
0	-1.22	1	-1.485	-0.265
1	1.5	3	1.615	0.115
2	3.24	4	3.165	-0.075
3	3.94	4	3.165	-0.775
4	2.20	3	1.615	-0.585
5	-1.10	1	-1.485	-0.385
6	-2.26	1	-1.485	0.775
7	-1.88	1	-1.485	0.395
8	-1.2	1	-1.485	-0.285

* The average quantized error power.

$$P_q = \frac{1}{N} \sum_{i=1}^N e_q^2 \quad \rightarrow \text{Number of Samples}$$

$N = 9$ in our problem.

$$\begin{aligned} P_q &= \frac{1}{9} [(-0.265)^2 + (0.115)^2 + (-0.075)^2 + (-0.775)^2 + (-0.585)^2 \\ &\quad + (-0.385)^2 + (0.775)^2 + (0.395)^2 + (-0.285)^2] \\ &= \frac{1}{9} [2.018025] = 0.224225. \end{aligned}$$

* Number of bits required for encoding:

$\rightarrow$ No. of bits
 $n_b = \log_2 L \rightarrow$ No. of levels.

$$= \log_2 4 = 2 \text{ Bits}$$

(00, 01, 10, 11) $\rightarrow$ Encoding for the 4 levels

5) Quantize and encode the following sampled $x(n)$ using 8 quantization levels and minimum number of bits.

$$x(n) = \{ 0.387, 0.430, 0.478, 0.531, 0.590, 0.6561, 0.729, 0.81, 0.9, 1, 0.2 \}$$

- Maximum Amp = 1
Minimum Amp = 0.2

$$\Delta = \frac{\text{Max Amp} - \text{Min Amp}}{L} = \frac{1 - 0.2}{8} = \underline{0.1}$$

- The eight levels are

$$[[0.2, 0.3], [0.3, 0.4], [0.4, 0.5], [0.5, 0.6], [0.6, 0.7], [0.7, 0.8], [0.8, 0.9], [0.9, 1]]$$

- The mid point for the eight levels

$$[0.25, 0.35, 0.45, 0.55, 0.65, 0.75, 0.85, 0.95]$$

n	$x(n)$	Interval Index	$x_q(n)$	$x_q(n) - x(n)$
0	0.387	2	0.35	-0.037
1	0.430	3	0.45	0.02
2	0.478	3	0.45	-0.028
3	0.531	4	0.55	0.019
4	0.590	4	0.55	-0.04
5	0.6561	5	0.65	-0.0061
6	0.729	6	0.75	0.021
7	0.81	7	0.85	0.04
8	0.9	8	0.95	0.05
9	1	8	0.95	-0.05
10	0.2	1	0.25	0.05